

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date: 24 June 2026

Project Name: Visualization Tools for Electric Vehicle Charge and Range Analysis

Functional Requirements

FR No.	Functional Requirement	Sub Requirement
FR-1	Data Import	Import EV data from CSV/Excel Validate dataset
FR-2	Data Preprocessing	Remove null values & duplicates Create calculated fields
FR-3	Data Visualization	Charging station, EV brand, efficiency & treemap dashboards Interactive filters
FR-4	Insights & Reporting	Generate Tableau Story Display key insights

Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Easy-to-use interactive dashboards
NFR-2	Security	Secure local database access
NFR-3	Reliability	Accurate preprocessing and visualization
NFR-4	Performance	Fast SQL queries and dashboard loading
NFR-5	Availability	Accessible on local system whenever required
NFR-6	Scalability	Supports additional EV datasets and dashboards